MCP + RAG for 1C — one-page summary

Local-first search over BSL business logic and configuration metadata · Sergey Matyukhin

PROBLEM

- A 1C configuration dump is thousands of .bsl modules plus a multi-megabyte metadata report; grep finds text, not meaning.
- An exact handler name and an intent question need different search strategies, and an LLM agent needs a tool with measured quality.
- Hosted search stacks require deployment work and send the code off the machine.

APPROACH

- Local MCP server over stdio: 9 tools, including an explainable router and a diagnostic index_stats.
- intfloat/multilingual-e5-small (384-d, 512-token limit) + Chroma with HNSW cosine, CPU only.
- Custom global BM25 inverted index; Reciprocal Rank Fusion with k = 60; optional flashrank cross-encoder, off by default.
- No external APIs and no data leaving the machine; 40-60 ms per query.

CORPUS

- 154,560 BSL code chunks (one procedure or function per chunk, ADR 0007) + 25,026 metadata chunks from the configuration XML = 179,586 chunks.
- Chunking gates: 99.6% code coverage; 0.3% of chunks over 512 tokens; token p50/p99/max = 179 / 472 / 554.
- Metadata source change (text report → XML): BM25 Hit@3 93.0% → 99.7% (ADR 0008).

QUALITY — GOLD-SET V2: 70 NATURAL-LANGUAGE QUERIES, FILE-LEVEL RELEVANCE, 4 RUNS, MEAN ± STD

Configuration	nDCG@10	Recall@10	Hit@1	MRR@10
vector	0.339 ± 0.023	0.523 ± 0.030	0.150 ± 0.007	0.268 ± 0.019
pearson	0.337 ± 0.030	0.511 ± 0.048	0.179 ± 0.021	0.270 ± 0.024
hybrid (RRF)	0.301 ± 0.017	0.482 ± 0.034	0.129 ± 0.010	0.240 ± 0.013
hybrid + rerank	0.117	0.157	0.086	0.102

Paired test (v2): Δ nDCG@10 = +0.037, bootstrap 95% CI [-0.015; +0.086], p = 0.154 → NOT significant; default unchanged, ADR 0009 proposed.
Independent v3 hold-out (71 code queries, 3 runs): vector 0.205 ± 0.014 vs hybrid 0.183 ± 0.005 nDCG@10; Δ +0.023, CI [-0.019; +0.061], p = 0.275 → also NOT significant.
Router on v3: 58/100; the main gap is lexical→hybrid. Lexical anchors: 10/10 in top-3. Cross-encoder stays optional (0.117).

ENGINEERING

- 72 unit tests; ruff clean, mypy over 22 modules, cyclomatic-complexity gate 143 blocks / avg 4.37 (A) in CI; Python 3.11/3.12; CI green at 64a0085 (cleaned history).
- ADR log 0001-0009 and CHANGELOG; router 68/68 on the dev set, 58/100 on the independent v3 hold-out (quoted-name lexical cases fall back to hybrid).
- Limitations: HNSW rebuilt on start (~60-70 s cold start, workaround for chroma#7090); single writer; no incremental indexing; gold set partly synthetic.

LINKS AND CONTACT

github.com/Smaturus/1c-mcp-rag-search · smaturus@gmail.com · MCP Python SDK · Chroma · multilingual-e5-small · BM25 · RRF · flashrank

All numbers come from the repository reports: gold-set v2 (4 runs) and independent v3 hold-out (3 runs).